Program Name : Computer Engineering

Academic Year : 2025 – 2026

Program Code : CO

Scheme : K Semester : VI	Course Name : Capstone Project Course Code : CPE - 316004

CAPSTONE PROJECT PROPOSAL & WORK OUTLINE

Title :	Mobile Shop Lifecycle Management System

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Introduction:

In today's rapidly growing digital marketplace, mobile phones have become an essential part of daily life. However, the lifecycle management of mobile devices—from purchase to warranty tracking, repair management, and resale—is often handled manually or through disconnected systems. Existing Point of Sale (POS) systems mainly focus on billing and inventory, while manufacturer websites only provide limited warranty checking features. Repair centers usually maintain isolated records that are not shared across different shops or stakeholders.

To overcome these limitations, the proposed Mobile Shop Lifecycle Management System will be developed as a centralized web-based platform. The system will manage the complete lifecycle of a mobile device using IMEI-based unique identification. It will integrate shop owners, service centers, customers, technicians, and administrators into a single digital ecosystem.

The system will ensure transparency, prevent fraud through duplicate IMEI detection, and maintain a permanent digital history of every registered mobile device. By leveraging modern web technologies, the system will enhance operational efficiency, improve customer trust, and streamline lifecycle tracking from purchase to resale.

Problem Statement:

Currently, mobile shops and service centers manage inventory, sales, warranty, and repair records separately. Most systems are limited to billing or stock management, and there is no centralized platform that tracks a mobile phone's complete lifecycle.

The major problems identified include the lack of centralized IMEI-based tracking, which makes it difficult to monitor devices efficiently across the ecosystem. There is no cross-shop visibility of repair history, preventing service providers from accessing complete and accurate device records. Warranty tracking processes are largely manual, leading to errors, delays, and inefficiencies. The system is also exposed to fraud risks due to duplicate IMEI usage, as there is no reliable mechanism to detect or prevent such duplication. Additionally, there is no transparent ownership transfer process during resale, creating trust and verification challenges between buyers and sellers. Overall, data remains scattered across multiple stakeholders, resulting in poor coordination, limited accountability, and reduced operational efficiency

Because of these limitations, customers do not have full transparency regarding repair history or warranty status. Shops and service centers cannot access complete lifecycle information, leading to inefficiencies and trust issues.Therefore, a structured and centralized web-based system will be developed to address these challenges effectively.

Objectives of the Project:

The main objectives of the proposed system are to develop a centralized IMEI-based mobile lifecycle tracking platform that enables complete device monitoring from sale to resale. It aims to implement a secure multi-role login system for Admin, Shop Owner, Service Center, Customer, and Technician to ensure controlled access and accountability. The system will include a robust mobile inventory and sales management module, along with automated warranty period management and expiry tracking to reduce manual errors. It will support comprehensive repair history logging and parts replacement tracking, while also providing cross-shop visibility of repair records to ensure transparency. Additionally, the platform will enable secure resale ownership transfers while retaining the device's full history. Fraud detection mechanisms will be integrated through duplicate IMEI verification. The system will also offer digital invoice storage with downloadable records and generate detailed sales, repair, and warranty analytics reports. Overall, the solution is designed to ensure transparency, accountability, and improved operational efficiency across all stakeholders.

Scope of the Project

The proposed Mobile Shop Lifecycle Management System will be developed as a centralized, web-based application that will manage the complete lifecycle of mobile devices from initial registration to resale. The system will primarily focus on IMEI-based tracking, ensuring that each mobile device is uniquely identified and its complete history is securely maintained in a centralized database. The application will allow shop owners to manage inventory, record sales transactions, assign buyers, and initiate warranty periods. Service centers will be able to access mobile details using IMEI and log repair history, including parts replacement and service costs. Customers will be provided with a dashboard where they will be able to view purchased mobiles, check warranty status, access repair history, and download invoices. The system will also support resale ownership transfer while retaining the complete lifecycle history to ensure transparency and prevent fraudulent activities. Administrative users will monitor system activities, approve shop and service center registrations, generate analytical reports, and manage user access. The project scope will be limited to a web-based implementation using Next.js and MongoDB, and it will not include direct integration with telecom providers or manufacturer databases for IMEI validation. The system will be designed in a scalable manner so that it can be extended to multiple shops and service centers in the future.

Proposed System / Methodology:

The proposed system will be implemented as a full-stack web application using Next.js for frontend and backend development and MongoDB as the primary database. The development methodology will follow a structured software development lifecycle approach, beginning with requirement analysis, followed by system design, implementation, testing, and deployment. The system will adopt a role-based access control mechanism where users such as Admin, Shop Owner, Service Center, Technician, and Customer will log in through secure authentication using JWT-based session management.

The application will function through a centralized IMEI-based workflow. When a shop owner registers a mobile device, the system will validate the uniqueness of the IMEI to prevent duplication and fraud. Once the mobile is sold, buyer details will be assigned, and the warranty lifecycle will automatically begin. Service centers will search devices using IMEI and record repair entries, including repair status, replaced components, and cost details. These entries will be stored in MongoDB and will be visible across authorized stakeholders, ensuring cross-shop repair history transparency. Customers will access their dashboard to monitor warranty status, repair timeline, and ownership history in real time.

The system will also implement automated warranty expiry tracking and notification mechanisms to inform users about upcoming expirations or repair updates. Analytical modules will generate sales reports, warranty reports, and repair statistics using data visualization tools integrated into the Next.js frontend. Throughout development, emphasis will be placed on data integrity, security, scalability, and performance optimization. The final system will provide a centralized, transparent, and efficient digital ecosystem for managing the complete lifecycle of mobile devices.

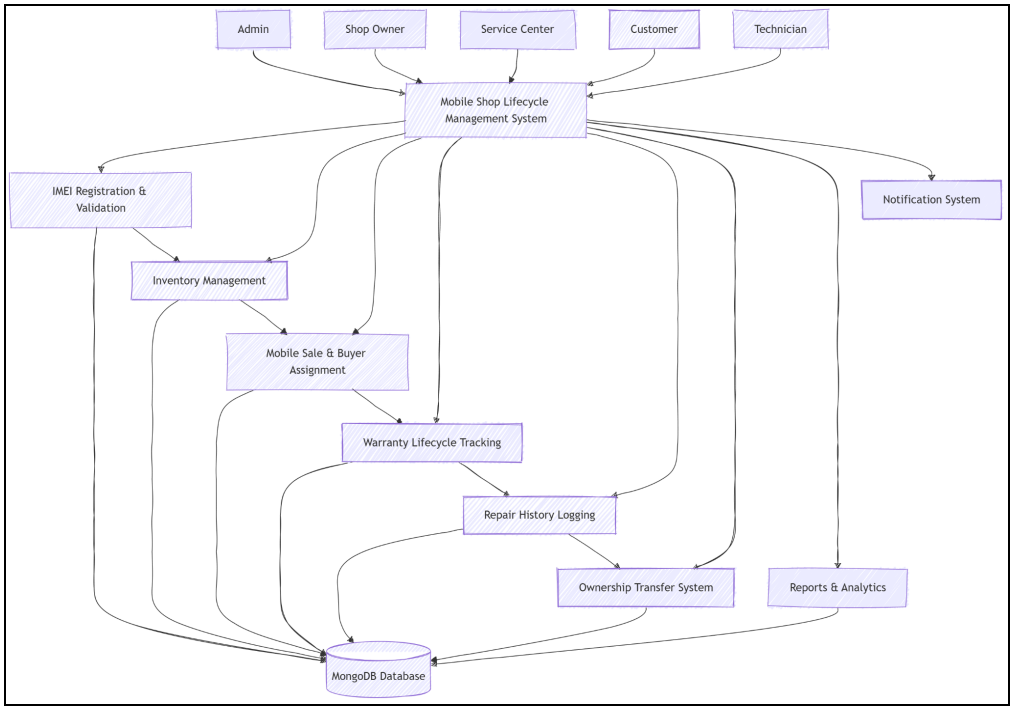

Fig 5.1:- Overall Mobile Shop Lifecycle Management System

Tools & Technologies:

- **Programming Language(s)**: Next.JS(backend), HTML, CSS,Next.JS (frontend)
- **Frontend / Backend/Database:**

  Frontend: Next.js, React.js, HTML, CSS, Tailwind CSS

  Backend: Next.js API Routes (Node.js environment)

  Database: MongoDB(Inbuilt).
- **Development Tools:**

  IDE: Visual Studio Code

  Version Control: Git / GitHub
- **Libraries:**

  Mongoose (MongoDB ODM),JWT for Authentication, Recharts (Data Visualization, graphs & visualization), chart.js,CSS(tailwind), JavaScript /Dom manipulation, jspdf.
- **Operating System:**

  Windows

Project Planning & Timeline

Phase	Activity	Duration	Dates
1	Requirement Analysis	2 Weeks	15/2/2026 to 27/2/2026
2	System Design	2 Weeks	28/2/2026 to 10/03/2026
3	Coding	5 Weeks	11/03/2026 to 14/03/2026
4	Testing	2 Weeks	15/03/2026 to 28/03/2026
5	Documentation	2 Week	01/04/2026 to 14/04/2026
	Total	**13 weeks**	

Expected Outcomes

The proposed Mobile Shop Lifecycle Management System is expected to deliver a fully functional, centralized web-based lifecycle management platform that enables secure IMEI-based mobile registration and tracking. It will establish a transparent and permanent system for maintaining complete repair and ownership histories, ensuring that all device-related activities are accurately recorded. The system will automate warranty lifecycle monitoring and expiry tracking, reducing manual intervention and errors. Fraud prevention will be strengthened through duplicate IMEI detection mechanisms. A role-based secure login system will be implemented for Admin, Shop Owner, Service Center, Technician, and Customer to ensure controlled access and accountability. Additionally, the platform will provide a centralized reporting system for sales, repairs, and warranty analytics, along with digital invoice generation and secure storage. Overall, the system will improve operational efficiency, enhance transparency among stakeholders, and build greater trust between buyers, sellers, and service providers by maintaining accurate and tamper-resistant digital records.

Conclusion

The proposed Mobile Shop Lifecycle Management System will provide a centralized and transparent digital platform for managing the complete lifecycle of mobile devices. By integrating shop owners, service centers, customers, technicians, and administrators into a single ecosystem, the system will eliminate fragmented record-keeping and manual tracking processes. The implementation of IMEI-based lifecycle monitoring will ensure uniqueness, prevent fraud, and maintain permanent digital history from purchase to repair and resale. Automated warranty tracking, centralized repair logging, and resale ownership transfer will improve operational efficiency and customer confidence.

By leveraging Next.js and MongoDB, the system will ensure scalability, security, and high performance. Overall, the project will contribute toward modernization of mobile shop management practices and create a reliable digital infrastructure for lifecycle transparency and data-driven decision-making.

References

- Official documentation of Next.js – https://nextjs.org/docs
- Official documentation of MongoDB – https://www.mongodb.com/docs/
- Mongoose ODM Documentation – https://mongoosejs.com/docs/
- JSON Web Token (JWT) Documentation – https://jwt.io/introduction
- Chart.js Documentation – https://www.chartjs.org/docs/latest/
- Research articles related to lifecycle management systems and centralized digital ecosystems.
- Online tutorials and technical blogs related to full-stack web development using Next.js and MongoDB

Undertaking by students

We undertake,

1. To develop the proposed system that will fulfil the scope as mentioned in 4
2. To follow the methodology as mentioned in 5
3. To follow the schedule as mentioned in 7

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